

$$S_{nh} = u + \frac{a}{2}(2n-1)$$

$$v = u + at$$

$$S = ut + \frac{1}{2}at^2$$

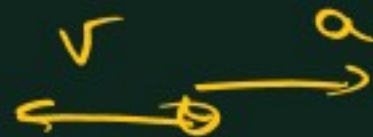
$$v^2 = u^2 + 2as$$

$$S = \left(\frac{u+v}{2} \right) t$$

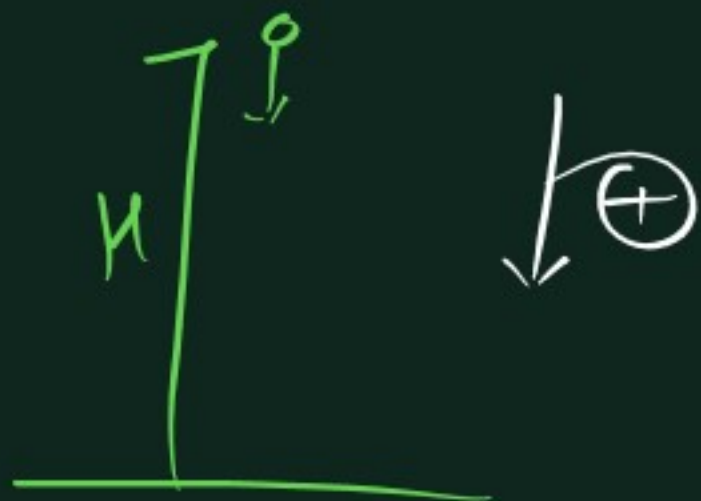
↳ v_{av}

$$\vec{v}_{av} = \frac{\vec{Disp}}{time}$$

$$Speed_{av} = \frac{Distance}{time}$$



Case I



$$a = +g$$

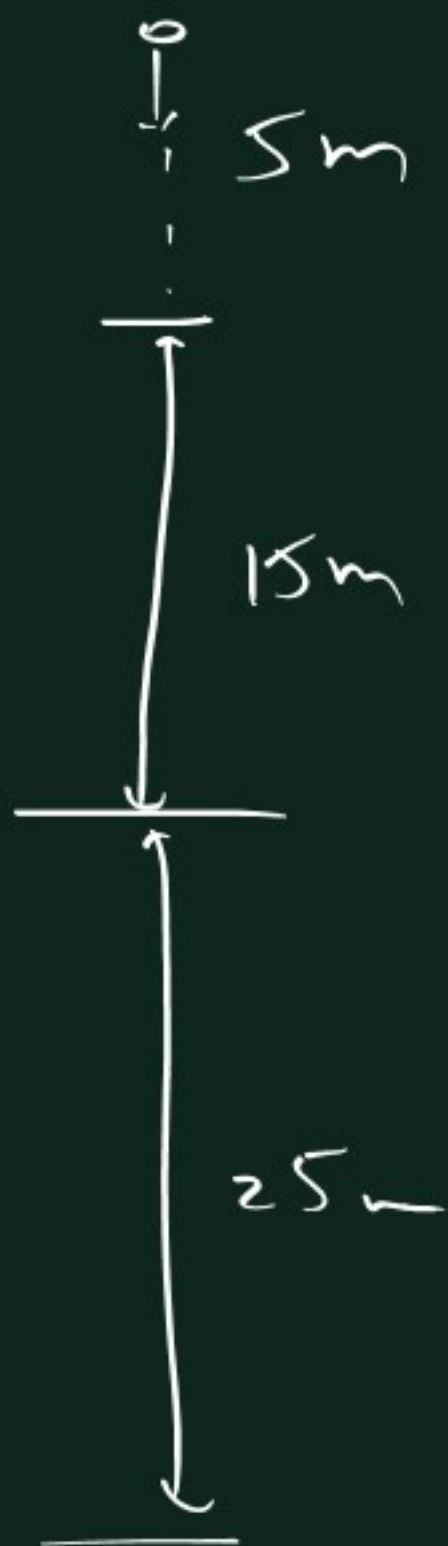
$$\Delta = +H$$

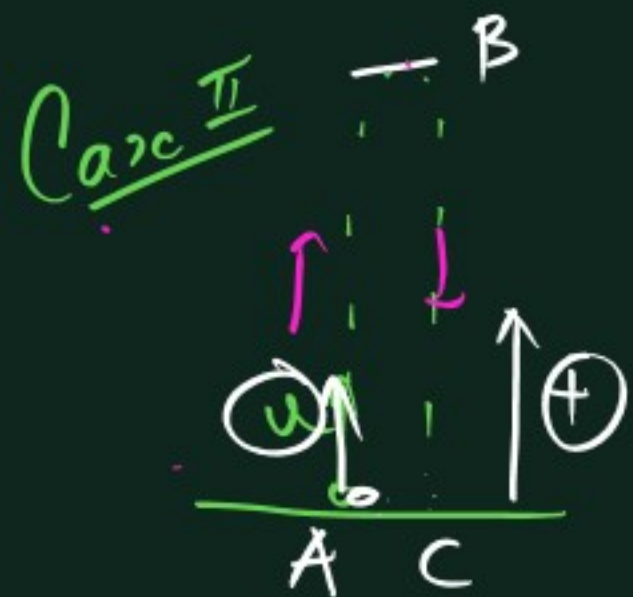
$$u = 0$$

$$t = \sqrt{\frac{2H}{g}}$$

$$v = \sqrt{2gH}$$

1st second





A to B

$$u = +u$$

$$a = -g$$

$$v = 0$$

$$s = ??$$

$$t = ??$$

$$v = u + at$$

$$\Rightarrow 0 = u - gt$$

$$\Rightarrow \underline{t = \frac{u}{g}}$$

$$v^2 = u^2 + 2as$$

$$\Rightarrow 0 = u^2 - 2gs$$

$$\Rightarrow \underline{s = \frac{u^2}{2g}}$$



$$\left[\begin{aligned} t_{\text{ascent}} &= \frac{u}{g} \\ t_{\text{descent}} &= \frac{u}{g} \end{aligned} \right]$$

$$\underline{h_{\text{max}} = \frac{u^2}{2g}}$$

$$\rightarrow \text{Total time, } \underline{T = \frac{2u}{g}}$$

Find total time & speed of landing

A to C

$$u = +u$$

$$s = 0$$

$$a = -g$$

$$s = ut + \frac{1}{2}at^2$$

$$\Rightarrow 0 = ut - g\frac{t^2}{2}$$

$$\Rightarrow t(u - \frac{gt}{2}) = 0$$

$$\Rightarrow \underline{t=0}, \underline{t = \frac{2u}{g}}$$

$$v = u + at$$

$$\Rightarrow v = u - g\left(\frac{2u}{g}\right)$$

$$\boxed{v = -u}$$



$$\left[v^2 = u^2 + 2as \right]$$

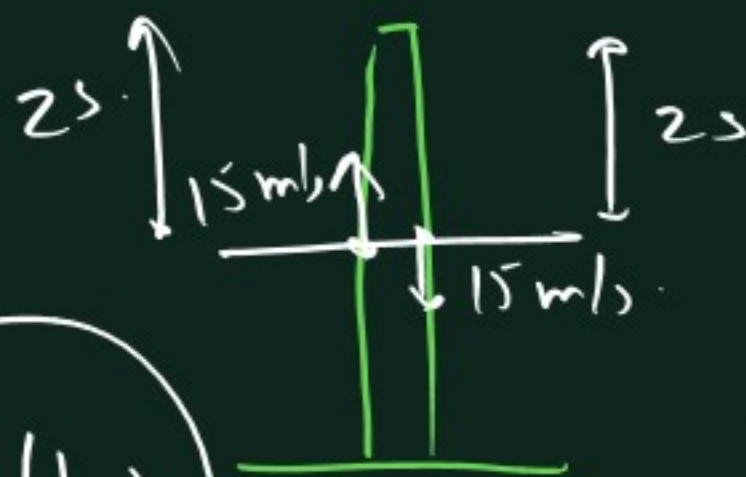
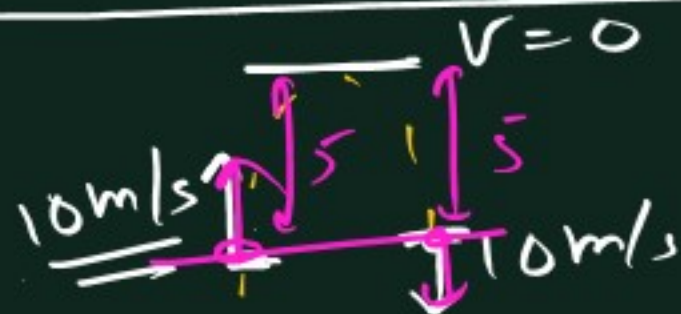
$$\Rightarrow v^2 = u^2$$

$$\Rightarrow v = \pm \sqrt{u^2}$$

$$\Rightarrow \underline{v = \pm u} \Rightarrow v = -u \quad (\downarrow)$$

$$x^2 = 4$$

$$\Rightarrow x = \pm \sqrt{4}$$



Q1

$$B \quad v=0$$



Find 1) $h_{\max}$ & time of ascent

2) distance covered during last second of ascent

3) the time(s) when height from ground is 20m.



$$u = +40$$

$$a = -10$$

$$v = 0$$

$$v^2 = u^2 + 2as$$

$$\Rightarrow 0 = 1600 - 2 \times 10 \times h$$

$$\Rightarrow h = \frac{1600}{20} = 80\text{m}$$

or $h = \frac{u^2}{2g} = \frac{40^2}{20} = 80\text{m}$

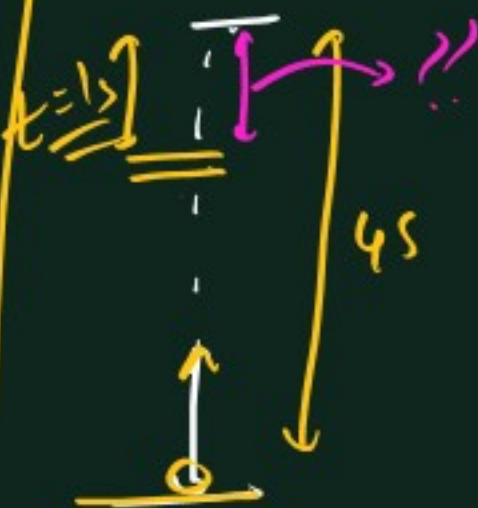
$$v = u + at$$

$$\Rightarrow 0 = 40 - 10t$$

$$\Rightarrow t = 4\text{s}$$

or $t_{\text{ascent}} = \frac{u}{g} = 4\text{s}$

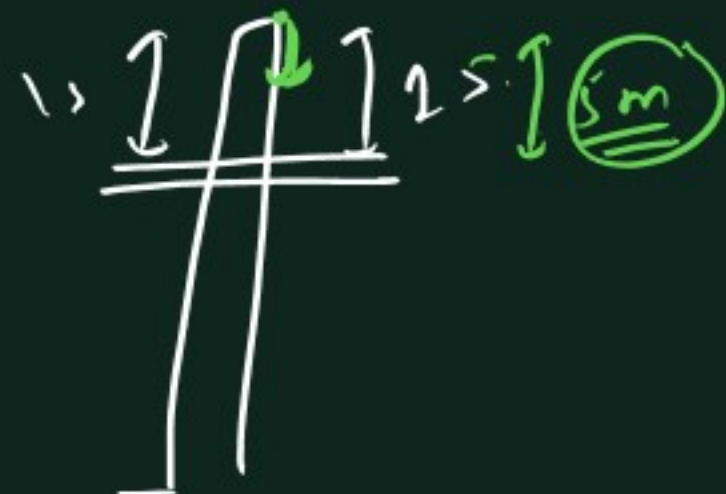
2) disp. in last second of ascent ??



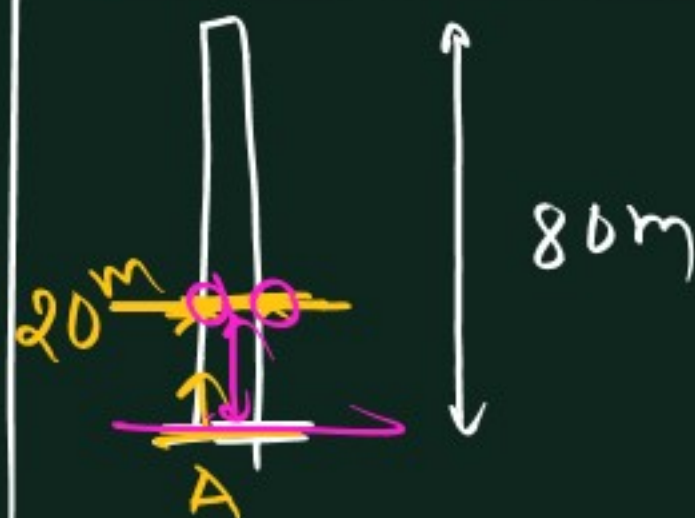
Displacement in 4th second

$$\begin{aligned} S_{4\text{th}} &= u + \frac{a}{2}(2 \times 4 - 1) \\ &= 40 - \frac{10}{2}(7) \\ &= 5\text{m} \end{aligned}$$

1) $v = 0$
 $v = 10\text{m/s}$
 $s = \frac{(u+v)}{2} \times t$
 $= 5\text{m}$



2nd way Symmetry



$$\begin{cases} u = +40 \\ s = +20 \\ a = -10 \end{cases}$$

$$t = ??$$

$$s = ut + \frac{1}{2}at^2$$

$$\Rightarrow 20 = 40t + \frac{1}{2}(-10)t^2$$

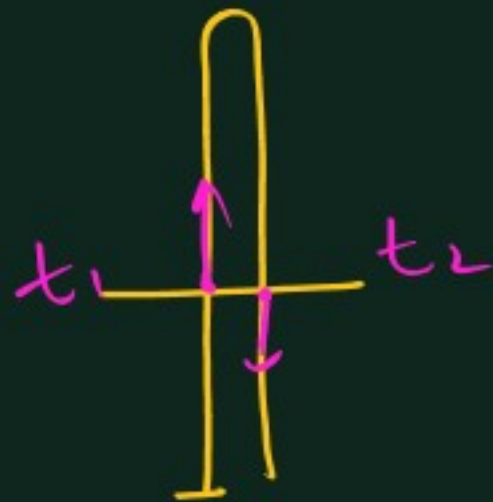
$$\Rightarrow 4 = 8t - t^2$$

$$\Rightarrow t^2 - 8t + 4 = 0$$

$$t = \frac{+8 \pm \sqrt{64 - 4 \times 4 \times 1}}{2}$$

$$= \frac{8 \pm \sqrt{48}}{2}$$

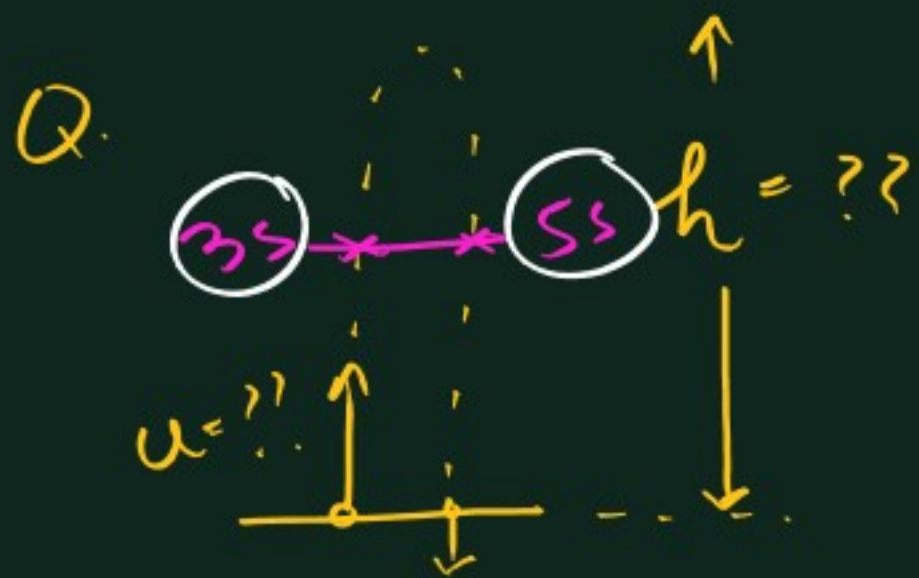
$$= \frac{8 \pm 4\sqrt{3}}{2} = 4 \pm 2\sqrt{3} = 4 \pm 3.46 = \underline{7.46, 0.54}$$



$$\text{Total time of flight} = \frac{2u}{g} = t_1 + t_2$$

$$(Add = 8s = \text{total time})$$

$$2 \times 1.73 = 3.46$$



$$t_1 + t_2 = T \text{ or } \frac{2u}{g}$$

$$\Rightarrow 8 = \frac{2 \times u}{10}$$

$$\Rightarrow \underline{u = 40 \text{ m/s}}$$

$$h = \frac{u^2}{2g} = \frac{40^2}{20} = \underline{\underline{80 \text{ m}}}$$



S & V_{avg} in 8 s

$$u = +60$$

$$a = -10$$

$$\underline{t = 8}$$

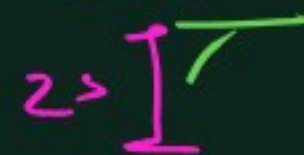
$$\textcircled{8} = ut + \frac{1}{2} at^2$$

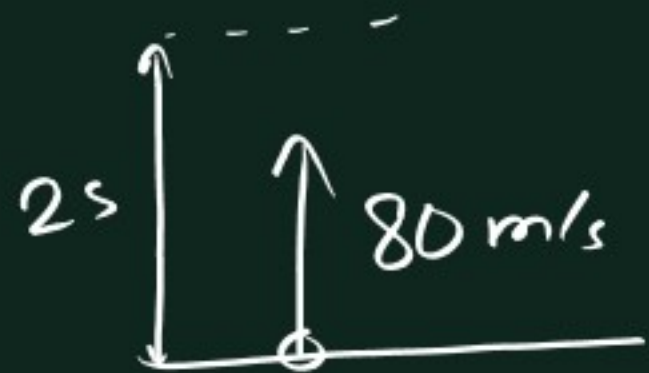
$$= 60 \times 8 - \frac{1}{2} \times 10 \times 8$$

$$= 480 - 320$$

$$\Rightarrow \underline{\underline{S = 160 \text{ m}}}$$

$$V_{\text{avg}} = \frac{S}{t} = \frac{160}{8} = \underline{\underline{20 \text{ m/s}}}$$





1st 2 seconds.

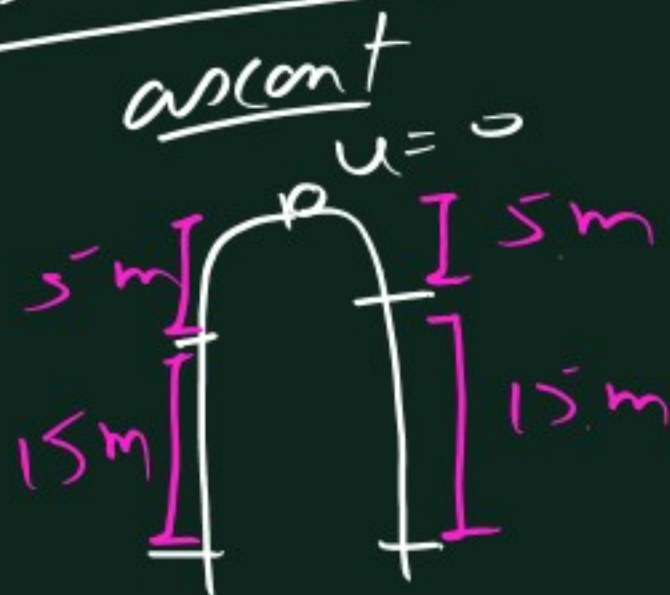
$$u = 80$$

$$a = -10$$

$$t = 2$$

$$\begin{aligned}
 S &= ut + \frac{1}{2}at^2 \\
 &= 80 \times 2 - \frac{10}{2} \times 2^2 \\
 &= \underline{140m}
 \end{aligned}$$

last 2 seconds of



20m in last
2 sec. of ascent

Ratio 140:20 = 7:1

2nd way

$$t_{\text{ascent}} = \frac{u}{g} = \underline{8s}$$

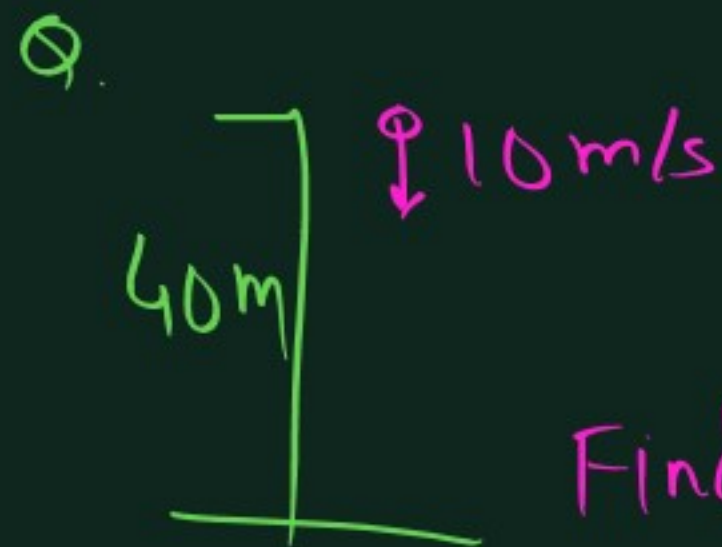
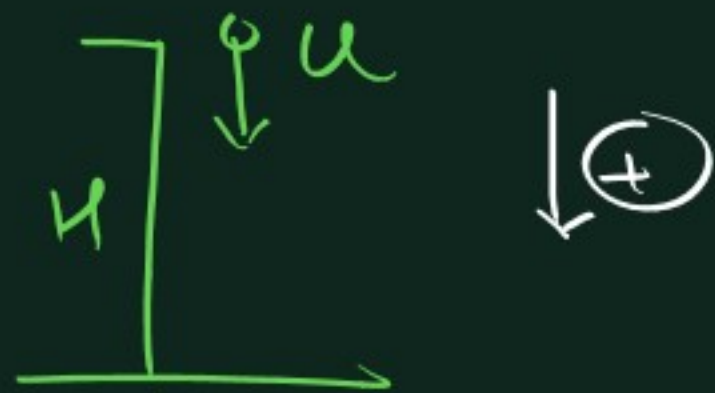
7th + 8th sec

$$\begin{aligned}
 &u + \frac{a}{2}(2 \times 7 - 1) \\
 &= 80 - \frac{10}{2}(13) \\
 &= \underline{15m}
 \end{aligned}$$

$$\begin{aligned}
 &80 - \frac{10}{2}(2 \times 8 - 1) \\
 &= \underline{5m}
 \end{aligned}$$

$$\text{Total} = \underline{20m}$$

Case III → Throwing down from a height



Find 1) speed of landing
2) time to land

$$u = +10$$

$$a = +10$$

$$s = +40$$

$$v^2 = u^2 + 2as$$

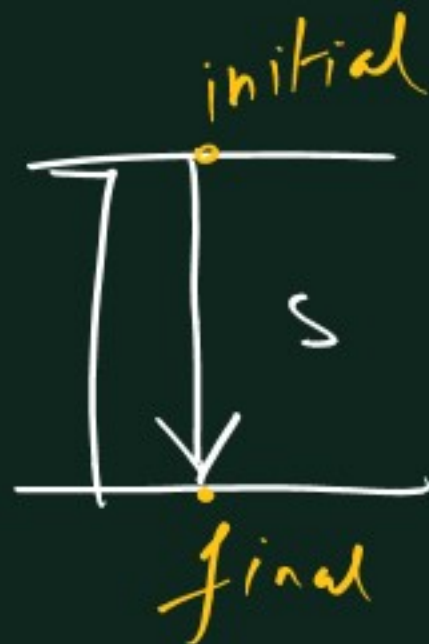
$$\Rightarrow v = \sqrt{u^2 + 2as}$$

$$= \sqrt{100 + 2 \times 10 \times 40}$$

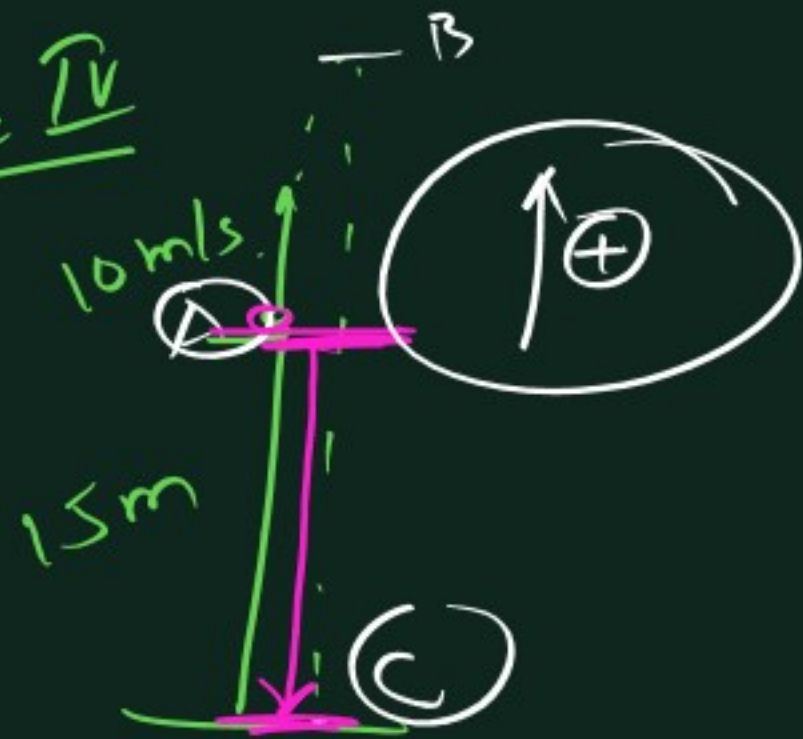
$$= \sqrt{900} = 30\text{ m/s}$$

$$v = u + at$$

$$\Rightarrow 30 = 10 + 10t \Rightarrow t = 2\text{ sec.}$$



Qax IV



Find time of flight & speed of landing



$$\begin{cases} u = +10 \\ a = -10 \\ \Delta = -15\text{m} \end{cases}$$

$$\Delta = ut + \frac{1}{2}at^2$$

$$\Rightarrow -15 = 10t + \frac{1}{2}(-10)t^2$$

$$\Rightarrow 5t^2 - 10t - 15 = 0$$

$$\Rightarrow t^2 - 2t - 3 = 0$$

$$\Rightarrow (t-3)(t+1) = 0$$

$$\Rightarrow t = 3, -1$$

↓

$$\therefore \text{time} = 3\text{s}$$

$$v = u + at$$

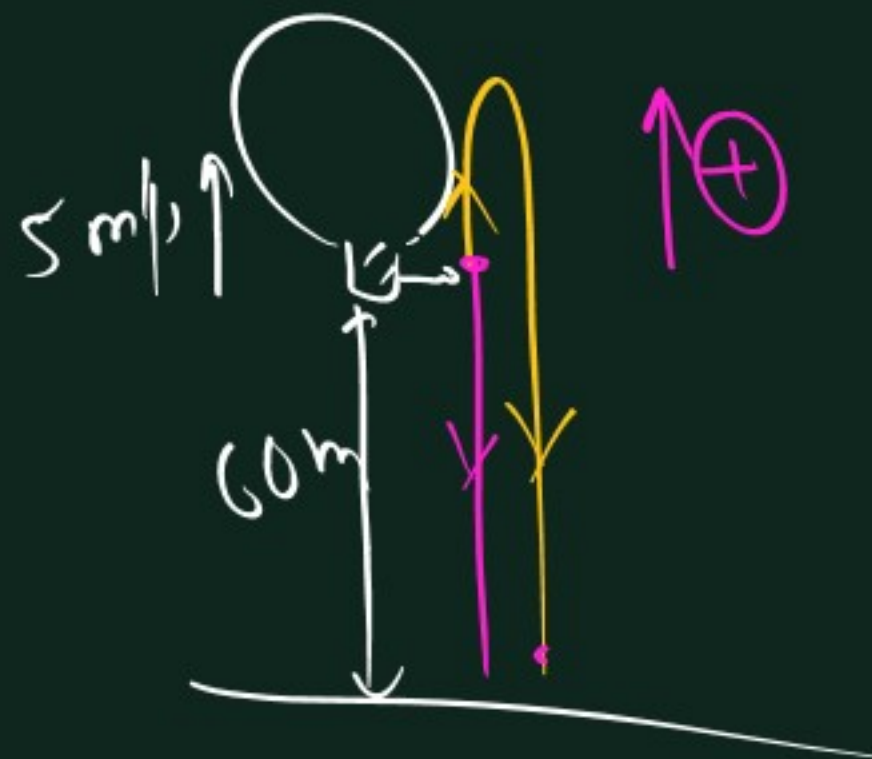
$$\Rightarrow v = 10 - 10 \times 3$$

$$= -20 \text{ (down)}$$



1) Time for stone to land.

2) height of balloon from ground when stone lands



$$u = +5$$

$$a = -10$$

$$s = -60$$

$$s = ut + \frac{1}{2}at^2$$

$$\Rightarrow -60 = 5t + \frac{1}{2}(-10)t^2$$

$$\Rightarrow -60 = 5t - 5t^2$$

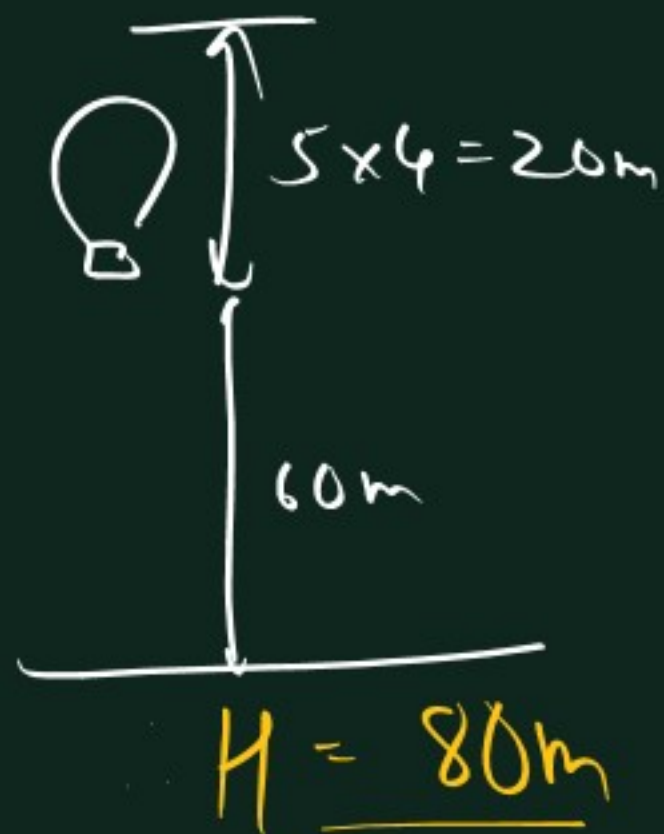
$$\Rightarrow -12 = t - t^2$$

$$\Rightarrow t^2 - t - 12 = 0$$

$$(t-4)(t+3) = 0$$

$$t = -3, 4$$

$$t = 4s$$





- Find out
- 1) time of ascent
 - 2) Max. height
 - 3) Distance in 3 sec.
 - 4) Avg. speed in 3 sec.

$$1) t_{\text{ascent}} = \frac{u}{g} = 2 \text{ s.}$$

$$2) h = \frac{u^2}{2g} = \frac{20^2}{20} = 20 \text{ m}$$

$$3) s = ut + \frac{1}{2} at^2$$

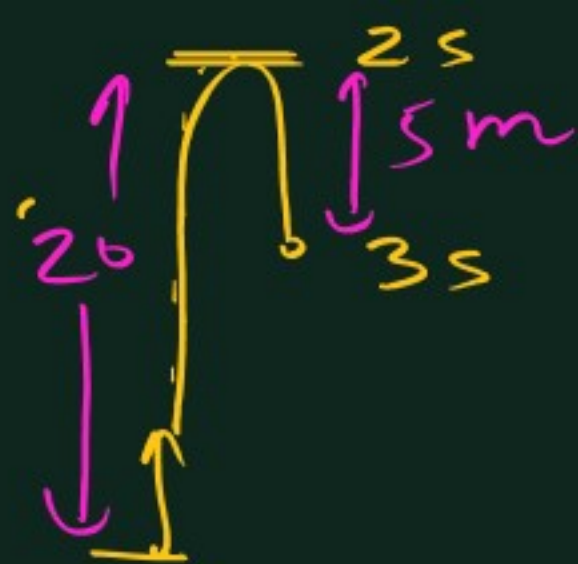
$$u = +20, a = -10, t = 3.$$

$$s = 20 \times 3 - \frac{1}{2} \times 10 \times 3^2$$

$$= 15 \text{ m}$$

↓
Displacement

⊗



$$\text{Distance } 20 + 5 = \underline{\underline{25 \text{ m}}}$$

$$4) \text{ Avg. speed} = \frac{\text{Distance}}{\text{time}} = \frac{25}{3}$$

4. Two bodies of different masses m_a and m_b are dropped from two different heights a and b . The ratio of the time taken by the two to cover these distances are [NCERT 1972; MP PMT 1993]

- (a) $a:b$ (b) $b:a$
(c) $\sqrt{a}:\sqrt{b}$ (d) $a^2:b^2$

5. A body falls freely from rest. It covers as much distance in the last second of its motion as covered in the first three seconds. The body has fallen for a time of [MNR 1998]

- (a) 3 s (b) 5 s
(c) 7 s (d) 9 s

12. A stone dropped from the top of the tower touches the ground in 4 sec. The height of the tower is about [MP PET 1986; AFMC 1994; CPMT 1997; BHU 1998; DPMT 1999; RPET 1999; MH CET 2003]

- (a) 80 m (b) 40 m
(c) 20 m (d) 160 m

19. A stone thrown upward with a speed u from the top of the tower reaches the ground with a velocity $3u$. The height of the tower is [EAMCET 1983; RPET 2003]

- (a) $3u^2/g$ (b) $4u^2/g$
(c) $6u^2/g$ (d) $9u^2/g$



$$t = \sqrt{\frac{2h}{g}}$$

$$\Rightarrow t^2 = \frac{2h}{g}$$

$$\Rightarrow h = \frac{gt^2}{2}$$

$$= \frac{10 \times 16}{2}$$

$$= 80m$$

13. A body is released from the top of a tower of height h . It takes t sec to reach the ground. Where will be the ball after time $t/2$ sec [NCERT 1981; MP PMT 2004]

- (a) At $h/2$ from the ground
(b) At $h/4$ from the ground
(c) Depends upon mass and volume of the body
(d) At $3h/4$ from the ground

42. A stone is shot straight upward with a speed of 20 m/sec from a tower 200 m high. The speed with which it strikes the ground is approximately [AMU (Engg.) 1999]

- (a) 60 m/sec (b) 65 m/sec
(c) 70 m/sec (d) 75 m/sec

40. A ball is dropped downwards. After 1 second another ball is dropped downwards from the same point. What is the distance between them after 3 seconds [BHU 1998]

- (a) 25 m (b) 20 m from start
(c) 50 m (d) 9.8 m

55. A particle when thrown, moves such that it passes from same height at 2 and 10s, the height is [UPSEAT 2000]

- (a) g (b) $2g$
(c) $5g$ (d) $10g$

1) $t = \sqrt{\frac{2h}{g}}$
 $t \propto \sqrt{h}$

$$\frac{t_1}{t_2} = \frac{\sqrt{a}}{\sqrt{b}}$$

2) ϕ in 3 seconds
 $=$ 5 in last sec.

$u = 0$
 $a = 10$
 $t = 3$
 $S = \frac{1}{2} a t^2 = \frac{1}{2} \times 10 \times 3^2 = 45m$

total time = n sec.

$$S_{nh} = 0 + \frac{a}{2} (2n-1)$$

$$= \frac{10}{2} (2n-1)$$

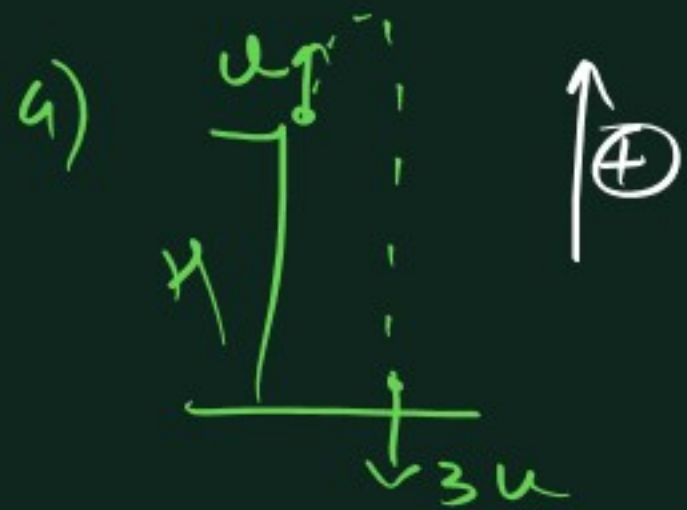
$$= \underline{5(2n-1)}$$

Given $45 = 5(2n-1)$
 $\Rightarrow n = 5$

2

5
15
25
35
45 } 45m

5th sec } 45-
 $t = 5s$



$$u = +u$$

$$v = -3u$$

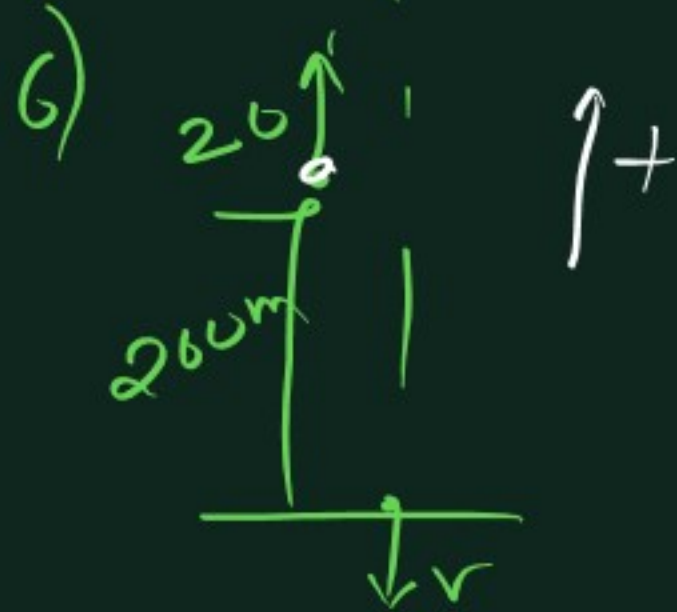
$$s = -H$$

$$a = -g$$

$$v' = u' + 2a$$

$$\Rightarrow 9u' = u' + 2(-g)(-H)$$

$$\Rightarrow 8u' = 2gH \Rightarrow H = \frac{4u'^2}{g}$$



$$u = +20$$

$$s = -200(\downarrow)$$

$$a = -10(\downarrow)$$

$$v^2 = u^2 + 2as$$

$$= 400 + 2 \times 10 \times 200$$

$$\Rightarrow v = -\sqrt{4400}$$

$$10 \sqrt{44} = 20 \frac{\sqrt{11}}{\sqrt{5}}$$

$$\sqrt{36} \cdot \sqrt{49}$$

$$6 \cdot 7$$

$$42$$

$$\sqrt{10} = 3.16$$

$$\sqrt{2} = 1.414$$

$$\sqrt{3} = 1.732$$

$$\sqrt{5} = 2.236$$

$$\sqrt{6} = 2.45$$

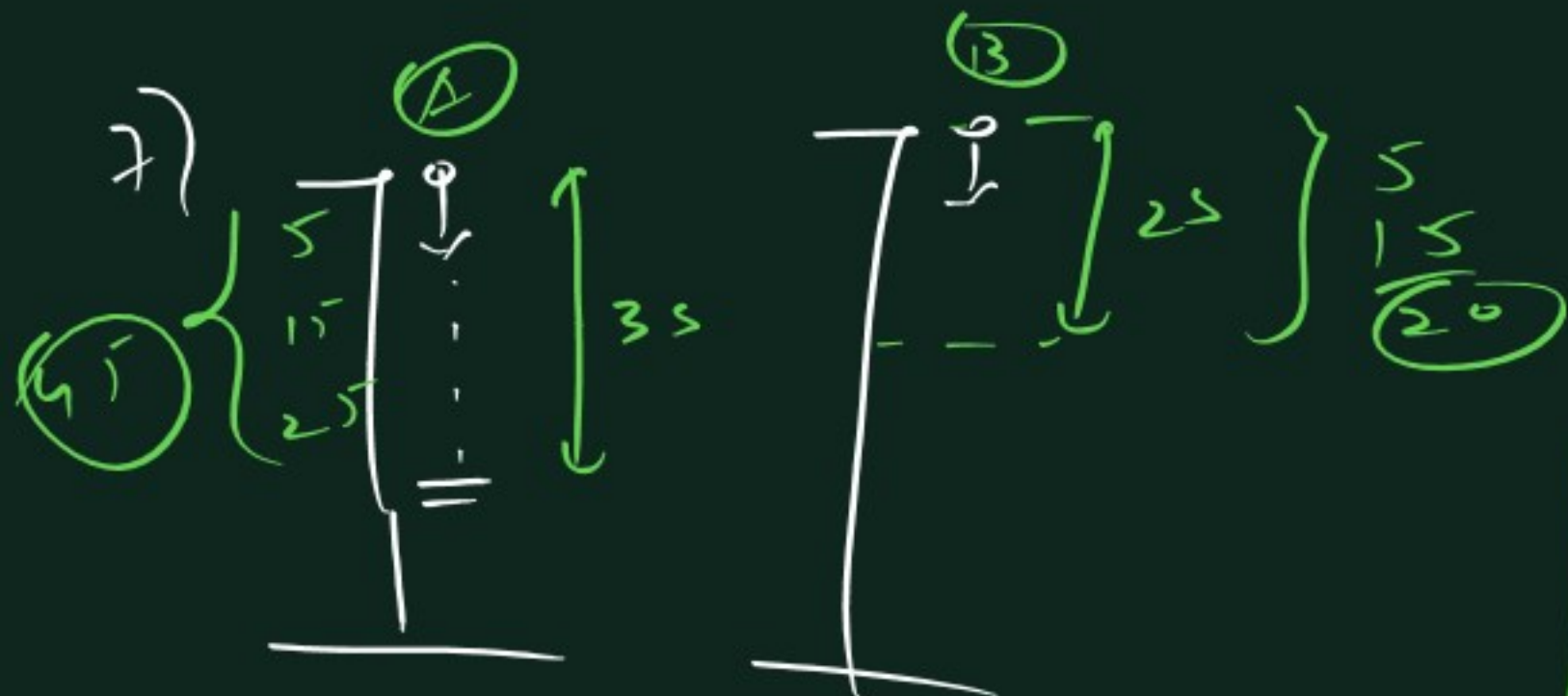
$$\sqrt{7} = 2.65$$

$$\sqrt{8} = 2\sqrt{2}$$

$$\sqrt{10} = 3.16 \approx \pi$$

$$\left(1 - \frac{20}{2}\right)^2$$

$$\left(1 - 10\right)^3$$



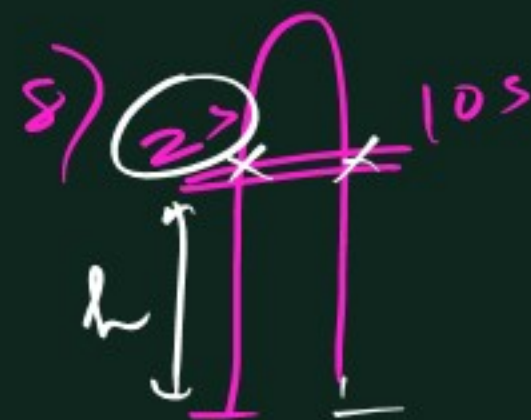
$$s = \frac{1}{2}at^2$$

$$= \frac{1}{2} \times 10 \times 3^2$$

$$= 45$$

$$s = \frac{1}{2}at^2$$

$$= 20$$



$$\text{Total } T = t_1 + t_2 = 12s$$

$$2\frac{u}{g} = 12$$

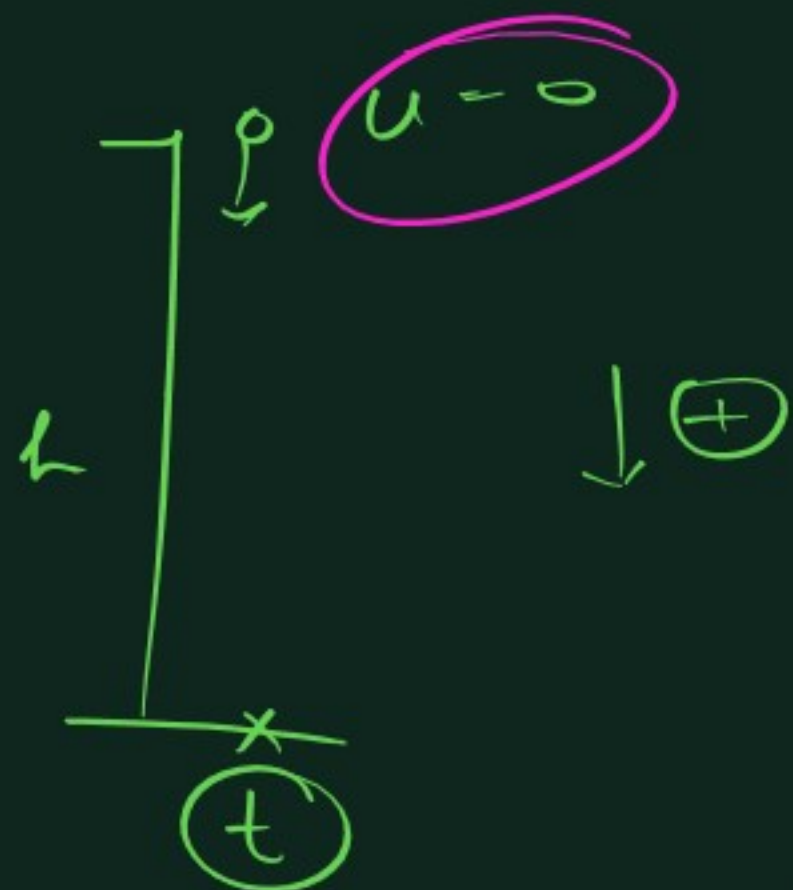
$$\Rightarrow u = \frac{12 \times 10}{2} = 60$$

$$s = ut + \frac{1}{2}at^2$$

(Put $t = 2s$ or $10s$)

$$s = 60 \times 2 - \frac{1}{2} \times 10 \times 2^2 = 100m$$

3)



(t)

$$s = ut + \frac{1}{2}at^2$$

$$\Rightarrow h = 0 + \frac{1}{2} \times g \times t^2$$

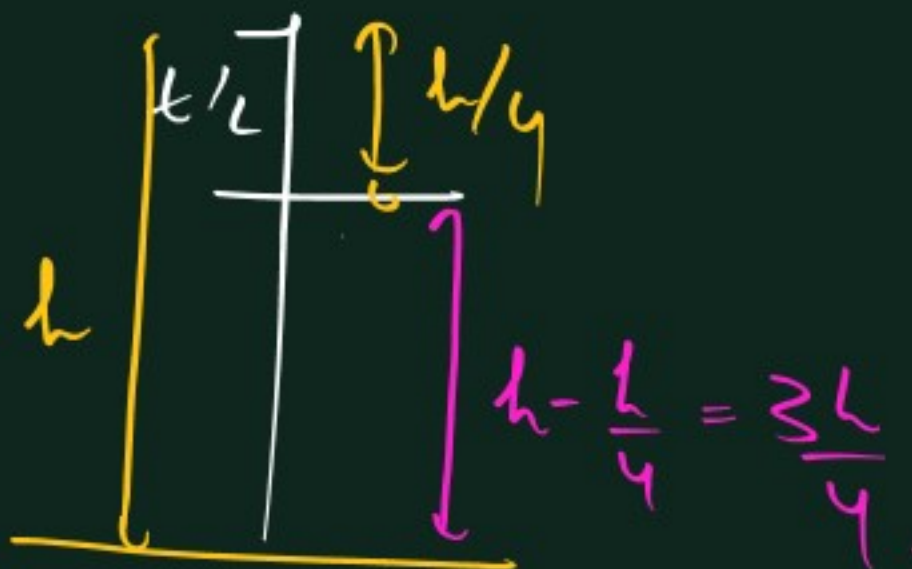
$$\Rightarrow h = \frac{1}{2}gt^2 \quad \text{--- (1)}$$

(t/2)

$$s = \frac{1}{2}at^2$$

$$= \frac{1}{2} \times g \times \left(\frac{t}{2}\right)^2$$

$$s = \frac{1}{2}g\frac{t^2}{4} = \frac{h}{4}$$



57. From the top of a tower, a particle is thrown vertically downwards with a velocity of 10 m/s . The ratio of the distances, covered by it in the 3^{rd} and 2^{nd} seconds of the motion is (Take $g = 10 \text{ m/s}^2$)

[AIIMS 2000; CBSE PMT 2002]

- (a) $5 : 7$ (b) $7 : 5$
(c) $3 : 6$ (d) $6 : 3$

72. A packet is dropped from a balloon which is going upwards with the velocity 12 m/s , the velocity of the packet after 2 seconds will be

[Pb PMT 2004]

- (a) -12 m/s (b) 12 m/s
(c) -7.6 m/s (d) 7.6 m/s

66. The acceleration due to gravity on the planet A is 9 times the acceleration due to gravity on planet B. A man jumps to a height of 2 m on the surface of A. What is the height of jump by the same person on the planet B [CBSE PMT 2003]

- (a) 18 m (b) 6 m
(c) $\frac{2}{3} \text{ m}$ (d) $\frac{2}{9} \text{ m}$

(A) $g \uparrow$ (B) $g \downarrow$

Diagram showing a particle falling from a tower with initial velocity $u = 10 \text{ m/s}$ downwards and acceleration $a = 10 \text{ m/s}^2$ downwards.

$$S_{3^{\text{rd}}} = u + \frac{a}{2} (2 \times 3 - 1)$$

$$= 10 + 5(5)$$

$$= 35$$

$$S_{2^{\text{nd}}} = 10 + \frac{10}{2} (2 \times 2 - 1)$$

$$= 25$$

Diagram showing a packet falling from a balloon with initial velocity $u = 12 \text{ m/s}$ upwards and acceleration $a = -10 \text{ m/s}^2$ downwards.

$u = 12$
 $a = -10$
 $t = 2$

$$v = u + at = 12 - 2 \times 10$$

Diagram showing a man jumping to a height $h = \frac{u^2}{2g} \Rightarrow h \propto \frac{1}{g}$.

$$\frac{h_1}{h_2} = \frac{g_2}{g_1} \Rightarrow \frac{2 \text{ m}}{h_2} = \frac{1}{9}$$

1. $0.4\hat{i} + 0.8\hat{j} + c\hat{k}$ represents a unit vector when c is

- (a) -0.2 (b) $\sqrt{0.2}$ (c) $\sqrt{0.8}$ (d) 0

3. The value of a unit vector in the direction of vector $A = 5\hat{i} - 12\hat{j}$, is

- 2) (a) $\hat{i}$ (b) $\hat{j}$ (c) $(\hat{i} + \hat{j})/13$ (d) $(5\hat{i} - 12\hat{j})/13$

6. Given that $\vec{A} + \vec{B} + \vec{C} = 0$ out of three vectors two are equal in magnitude and the magnitude of third vector is $\sqrt{2}$ times that of either of the two having equal magnitude. Then the angles between vectors are given by

- (a) $30^\circ, 60^\circ, 90^\circ$ (b) $45^\circ, 45^\circ, 90^\circ$ (c) $45^\circ, 60^\circ, 90^\circ$ (d) $90^\circ, 135^\circ, 135^\circ$

10. A car travels a distance S on a straight road in two hours and then returns to the starting point in the next three hours. Its average velocity is

- 4) (a) $S/5$ (b) $2S/5$ (c) $S/2 + S/3$ (d) None of the above

11. A particle moves along the sides AB, BC, CD of a square of side 25 m with a velocity of 15 ms^{-1} . Its average velocity is

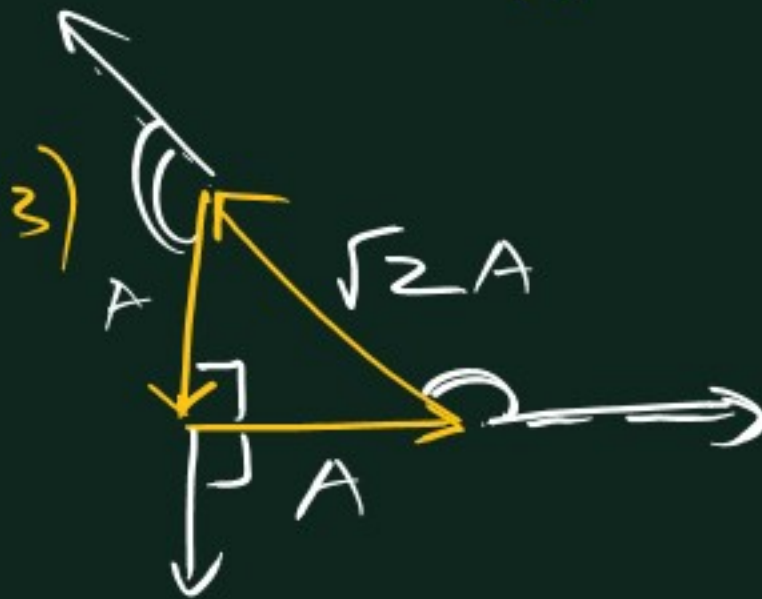
- 1) (a) 15 ms^{-1} (b) 10 ms^{-1} (c) 7.5 ms^{-1} (d) 5 ms^{-1}

12. A body has speed $V, 2V$ and $3V$ in first $1/3$ of distance S , second $1/3$ of S and third $1/3$ of S respectively. Its average speed will be

- 9) (a) V (b) $2V$ (c) $\frac{18}{11}V$ (d) $\frac{11}{18}V$

$$1) \sqrt{0.4^2 + 0.8^2 + c^2} = 1$$

$$2) \hat{A} = \frac{\vec{A}}{|\vec{A}|} = \frac{5\hat{i} - 12\hat{j}}{\sqrt{5^2 + 12^2}}$$



$$s = 25$$

$$\text{time} = \frac{\text{Distance}}{\text{Speed}}$$

$$= \frac{75}{15} = 5\text{ s}$$

$$\bar{V}_{\text{avg}} = \frac{s}{t} = 5$$

39. When a body is acted by a constant force, then which of the following quantities remains constant

- (a) Velocity (b) Acceleration (c) Momentum (d) None of these

40. A man of weight mg is moving up in a rocket with acceleration $4g$. The apparent weight of the man in the rocket is

- (a) Zero (b) $4mg$ (c) $5mg$ (d) mg

42. As shown in the figure, two equal masses each of 2 kg are suspended from a spring balance. The reading of the spring balance will be

- (a) Zero (b) 2 kg (c) 4 kg (d) Between zero and 2 kg

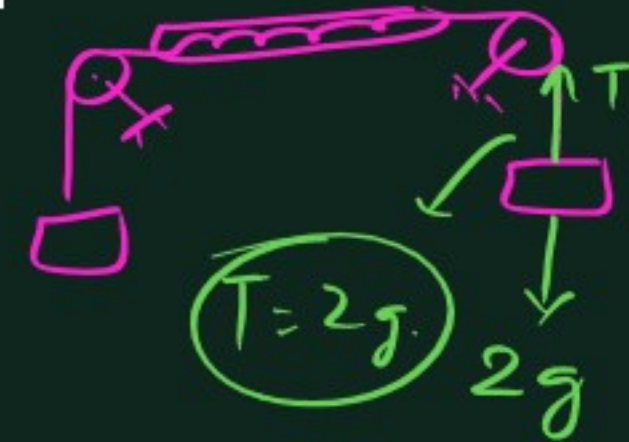
43. A player kicks a football of mass 0.5 kg and the football begins to move with a velocity of 10 m/s . If the contact between the leg and the football lasts for $\frac{1}{50}\text{ sec}$, then the force acted on the football should be

- (a) 2500 N (b) 1250 N (c) 250 N (d) 625 N

48. Two masses M and m are connected by a weightless string. They are pulled by a force F on a frictionless horizontal surface. The tension in the string will be

- (a) $\frac{FM}{m+M}$ (b) $\frac{F}{M+m}$ (c) $\frac{FM}{m}$ (d) $\frac{Fm}{M+m}$

Reading of machine



$$N - mg = ma$$

$$\Rightarrow N = mg + ma$$

$5mg$

$$10) F = \frac{P_2 - P_1}{t} = \frac{mv - m\cancel{u}}{t} = \frac{0.5 \times 10}{(1/50)}$$

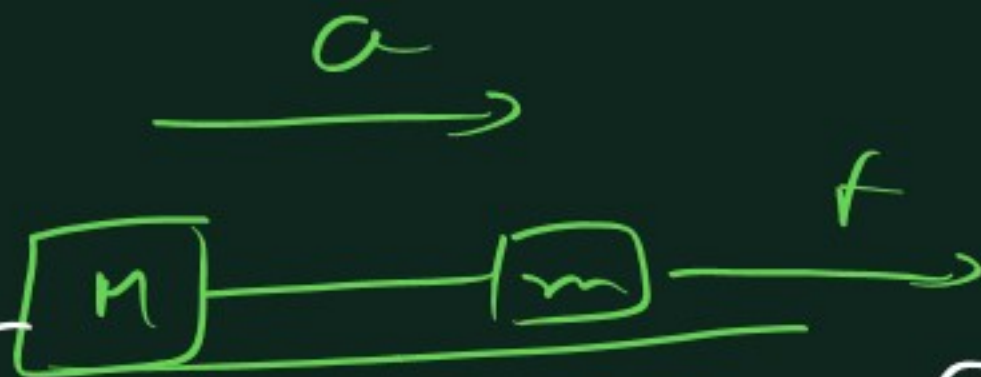
6)



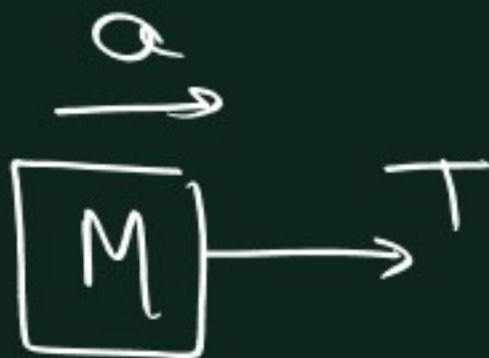
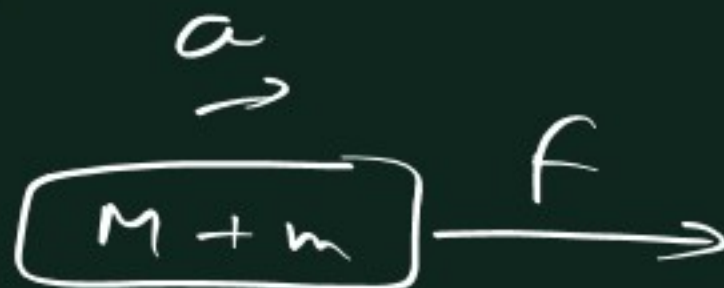
$$V_{\text{avg}} = \frac{S}{\left(\frac{S/3}{V}\right) + \left(\frac{S/3}{2V}\right) + \left(\frac{S/3}{3V}\right)}$$

$$= \frac{18V}{11}$$

11)



$$a = \frac{F}{M+m}$$



$$\Sigma F = Ma$$

$$\Rightarrow T = \frac{M \times F}{M+m}$$